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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 8

Software

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CHAPTER 8

Software

Computer Awareness

IN THIS CHAPTER

- System software
- Application software
- Language translators
- Utilities and debuggers
- Software licensing
- Booting

- CPU Socket
- Memory Slots
- Expansion Slots
- CMOS Battery
- SMPS

  - Ports:

- USB → Mouse, Keyboard
- HDMI → Audio + Video
- VGA → Monitor
-→ PS/2 Keyboard & Mouse
-→ Ethernet LAN
-→ SATA Hard Disk
-→ AGP Graphics Card
-→ CMOS Stores BIOS Settings
- SMPS → Power Supply Unit

SOFTWARE

Software is a collection of programs, instructions, and procedures that tell a computer what to do.

A Program is a set of instructions written in a programming language that directs the computer to perform a specific task.

Unlike hardware, software cannot be touched or physically seen. It exists in digital form and works with hardware to perform various operations.

Examples

- Windows 11

- Linux
- MS Word
- Google Chrome
- Adobe Photoshop
- VLC Media Player

CLASSIFICATION OF SOFTWARE

Software

- System Software
- Application Software
- Programming Software

1. SYSTEM SOFTWARE

System Software is the software that controls and manages the computer hardware and provides a platform for application software to run. A computer cannot function without system software.

Examples

- Windows
- Linux
- macOS
- Android
- BIOS
- Device Drivers

Functions of System Software

1. Controls Hardware

Manages CPU, memory, storage devices, and input/output devices.

2. Provides User Interface

Allows users to interact with the computer.

3. Runs Application Programs

Provides an environment for application software.

4. Manages System Resources

Controls:

- Memory
- Processor
- Storage

- Peripheral Devices

OPERATING SYSTEM (OS) An Operating System (OS) is system software that acts as an interface between the user and the computer hardware.

The operating system acts as a bridge between the user and the computer.

User

Operating System

Hardware

Functions of Operating System

- Memory Management
- File Management
- Device Management
- Process Management
- Security Management

Examples

- Microsoft Windows
- Linux
- macOS
- Android
- iOS

BIOS (Basic Input Output System)

BIOS is a system software stored in ROM (or Flash Memory in modern systems). It is the first software that runs when a computer is switched on.

Functions:

- Checks hardware components
- Loads boot program
- Starts operating system
- Initializes hardware devices

Location

Stored in:

- ROM
- Flash Memory

BOOT PROGRAM

A Boot Program is a small program that starts the operating system.

When a computer is powered on, BIOS loads the boot program into RAM.

The boot program then loads the operating system.

DEVICE DRIVER

A Device Driver is system software that enables the operating system to communicate with hardware devices.

Examples

- Printer Driver
- Scanner Driver
- Graphics Driver
- Audio Driver

LANGUAGE TRANSLATORS

A Translator is a software program that converts source code(Human readable language) into machine-readable code i.e object code (computer readable language).

Computers understand only 0 and 1 Therefore, programs written by humans must be translated into machine language(0 and 1).

Types of Translators

Translator

- Assembler
- Compiler
- Interpreter

1. Assembler

Converts Assembly Language into Machine Language.

2. Compiler

A Compiler translates the entire source code into object code at once.

Features

- Translates whole program
- A compiler can report multiple errors during translation; it is not guaranteed to find every error in one pass.
- Produces object code

Examples

- C Compiler

- C++ Compiler

Compiler = Complete Program

3. Interpreter

An Interpreter translates and executes one statement at a time.

Features

- Line-by-line translation
- Stops when error occurs
- No object code generated separately

Examples

- Python Interpreter
- JavaScript Interpreter

Interpreter = Individual Lines

UTILITY SOFTWARE

Utility Programs are small system programs that help maintain, manage, and optimize computer resources.

Also Called Service Programs

Types of Utility Programs

1. Disk Formatting

Prepares a storage device for use by creating a file system.

Example

- Formatting a Pen Drive before use.

2. Disk Cleanup

Removes unnecessary files to free disk space.

Removes

- Temporary files
- Cache files
- Recycle Bin contents

3. Disk Compression

Compresses data to store more information in less space.

4. Backup Program

Creates copies of files and data for recovery purposes.

Example

- Google Drive Backup

5. Debugger

A debugger helps locate and investigate program errors. A programmar changes the code to fix them.

Important Terms:

- Bug:An error in a program.
- Debugging:The process of finding and correcting bugs.
- Patch:A small software update used to fix bugs.

6. Disk Defragmentation

Rearranges fragmented files on a hard disk to improve performance.

Benefits

- Faster file access
- Improved efficiency

Note Primarily useful for HDDs, not SSDs.

7. File System

A File System is responsible for storing, organizing, retrieving, and managing files on storage devices.

Functions

- Create Files
- Delete Files
- Rename Files
- Modify Files
- Store Files

Examples

- FAT32
- NTFS
- exFAT

POST (Power-On Self-Test)

POST is a built-in diagnostic test performed when the computer starts.

Functions

Checks:

- RAM
- Keyboard
- Processor
- Storage Devices

Exam Point

POST checks hardware before booting.

BOOTING

Booting is the process of loading the operating system into memory(RAM) when the computer starts.

Types of Booting

1. Cold Boot

Starting a computer when it is completely powered off.

Example

Pressing the power button after the computer is shut down.

2. Warm Boot

Restarting a computer without switching off the power.

Example

Using Restart option.

2.APPLICATION SOFTWARE

Application Software is designed to perform specific tasks for end users.

A computer can run without application software, but it cannot perform user-specific tasks.

Examples

Productivity Software

- MS Word
- MS Excel
- MS PowerPoint

Multimedia Software

- VLC Media Player

Graphics Software

- Adobe Photoshop

Social Media Applications

- Facebook
- Instagram

Browsers

- Google Chrome
- Mozilla Firefox

Application Software Features

- User-oriented
- Task-specific
- Easy to use

IMPORTANT APPLICATION SOFTWARES

Operating System	Word Processing	Spreadsheet	Presentation	Database System
MS-DOS	Word Star	Lotus 1-2-3		D- base
MS-Windows	Word Pad Ms word	MS-Excel	Microsoft PowerPoint; PageMaker is desktop-publishing software.	M-access
Linux	K-word Ab-word Open writer,Star writer	K-spread Open calc Star calc	K-presents Star impress	Star base

SYSTEM SOFTWARE vs APPLICATION SOFTWARE

System Software	Application Software
Controls hardware	Performs user tasks
Necessary for computer operation	Optional
Example: Windows	Example: MS Word
Starts first	Starts after OS

3.PROGRAMMING SOFTWARE

Programming Software helps programmars create, edit, test, debug, and maintain other software.

Also Called

- Programming Tools
- Software Development Tools

Examples

Software	Purpose
Eclipse	Java Development
Coda	Mac Code Editor
Notepad++	Windows Code Editor
Sublime Text	Cross-platform Editor

SOFTWARE PIRACY

Software Piracy is the unauthorized copying, distribution, or use of software without permission from its owner.

TYPES OF SOFTWARE BASED ON LICENSING

1. Open Source Software

Software whose source code(programming code) is publicly available and can be modified.

Features

- Open-source software can be free of charge or sold; its licence provides source-code permissions.
- Customizable
- Community developed

Examples

- Linux
- Mozilla Firefox
- LibreOffice

2. Freeware

Software available free of cost, but source code is not available.

Examples

- Adobe Acrobat Reader
- Skype
- Google Chrome

3. Shareware

Software provided free for a limited trial period and later requires payment.

Examples

- ChatGpt

4. Nagware

Definition

Software that repeatedly reminds users to purchase the full version.

Features

- Trial version
- Frequent pop-up reminders

Open Source vs Freeware vs Shareware

Feature	Open Source	Freeware	Shareware
Free	Not necessarily; licence permissions concern source code	Yes	Temporary
Source Code Available	Yes	No	No
Modification Allowed	Yes	No	No
Example	Linux	Chrome	WinRAR

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

- Software is a collection of programs and instructions.
- Operating System is system software.
- BIOS is the first software to run.
- BIOS is stored in ROM.
- Device Driver controls hardware devices.
- Compiler translates the entire program.
- Interpreter translates line by line.
- Assembler converts assembly language into machine language.
- POST stands for Power-On Self-Test.
- Cold Boot starts a powered-off computer.
- Warm Boot means restart.
- A debugger helps locate and inspect bugs; it does not automatically remove all bugs.
- Software Piracy is illegal copying of software.
- Linux is open-source software.

ONE-LINE REVISION POINTS

- Software is a set of instructions.
- OS acts as an interface between user and hardware.
- BIOS starts the computer.
- Driver controls hardware.
- Compiler translates entire code.
- Interpreter translates line by line.
- Bug means error.

- A debugger helps investigate bugs so that the code can be corrected.
- POST checks hardware.
- Booting loads OS into memory.
- Linux is open source.
- Software piracy is illegal.

QUICK REVISION NOTES

Software

- System Software
- OS
- BIOS
- Drivers
- Utilities
- Application Software
- MS Word
- Photoshop
- VLC
- Programming Software
- Eclipse
- Notepad++
- Sublime Text

Translator:

Assembler → Assembly Language

Compiler → Entire Program

Interpreter → Line by Line

Booting:

Cold Boot = Power ON

Warm Boot = Restart

POST = Hardware Check

Linux = Open Source

COMPUTER MEMORY

A computer is an electronic device that accepts raw data (input), processes it according to a set of instructions, and produces meaningful information (output).

In simple words, a computer takes data from the user, stores it temporarily in memory, processes it using the CPU, and then provides the desired result.

WORKING OF A COMPUTER

INPUT → PROCESSING → OUTPUT

During processing, the computer stores both data and instructions in memory.

Example

Input: Student marks entered into a computer

Processing: Calculation of total marks and percentage

Output: Result displayed on the screen

Important Exam Point

A computer is also known as a Data Processing System because it converts raw data into useful information.

WHAT IS MEMORY?

Memory is the storage area of a computer where data, instructions, and information are stored.